Quantification: I534_z11c1-2
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PRE-VALIDATION: no number here has been checked against manual (ImageJ) counts.
PIXEL UNITS: lengths px, areas px^2, densities px^-1. The images carry no micron calibration.
Statistics are over interior lacunae only (those not touching the frame edge).

Per image
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measure	value
lacuna count	36
interior lacuna count	29
frame-edge lacuna count	7

Interior lacunae only: mean, median and sample SD (n = 29)
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measure	kind	mean	median	sd	units
area_px2	lacuna shape	2501.17	1735	2569.48	px^2
major_axis_length_px	lacuna shape	77.3811	74.0209	33.6741	px
minor_axis_length_px	lacuna shape	41.4006	36.7629	22.0227	px
aspect_ratio	lacuna shape	2.0208	1.9891	0.6173	unitless
eccentricity	lacuna shape	0.8055	0.8644	0.1715	unitless
solidity	lacuna shape	0.7378	0.7236	0.0923	unitless
roots_count	headline	10.3103	9	5.6259	count
ring_length_r30_px	headline	281.759	248	131.244	px
ring_length_r60_px	reported	519.897	483	261.815	px
owned_length_px	ownership-dependent	584.537	426.534	464.739	px
edge_count	ownership-dependent	29.1724	23	21.9497	count
mean_edge_length_px	ownership-dependent	20.0325	19.7165	5.7422	px
perimeter_px	normalised	253.534	236.936	125.571	px
ring_area_r30_px2	normalised	8504.72	7100	3445.07	px^2
ring_area_r60_px2	normalised	16527.5	13707	7001.15	px^2
in_frame_fraction_r30	normalised	0.9888	1	0.0308	unitless
in_frame_fraction_r60	normalised	0.9705	1	0.0682	unitless
ring_density_r30	normalised	0.0337116	0.0326977	0.0107046	px^-1
ring_density_r60	normalised	0.0319368	0.0345756	0.0103076	px^-1
roots_per_100px_perimeter	normalised	4.1779	4.137	1.3983	per 100 px
ring_attached_length_r30_px	network v2	255.345	221	134.117	px
ring_attached_length_r60_px	network v2	412.621	315	265.194	px
ring_length_w_r30_px	network v2	319.673	279.966	148.716	px
ring_length_w_r60_px	network v2	590.198	534.433	297.824	px
sholl_crossings_r10	network v2	6.2759	5	3.283	count
sholl_crossings_r20	network v2	7.1379	6	3.8053	count
sholl_crossings_r30	network v2	7.6897	7	4.2266	count
loop_count	loops, report only	1.3793	0	2.4555	count
loop_edge_share	loops, report only	0.099	0	0.1419	share
ring_width_mean_r30_px	width	7.832	7.6247	1.1642	px

Per field
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measure	value	units
analysed_area_px2	885683	px^2
total_skeleton_length_px	25214	px
canalicular_length_density_per_px	0.0284684	px^-1
junction_count	550	count
junction_density_per_px2	0.00062099	px^-2
lacuna_count	36	count

lacunae_per_px2	4.065e-05	px^-2
skeleton_component_count	314	count
mean_component_length_px	80.2994	px
median_component_length_px	23	px
field_length_density_w_per_px	0.0322797	px^-1
field_density_without_flagged_per_px	0.0298895	px^-1
field_density_in_roi_per_px		px^-1
unattached_length_px	184	px
unattached_component_count	12	count
canalicular_length_density_attached_per_px	0.0284684	px^-1
field_length_density_with_unattached_per_px	0.0286762	px^-1
loop_count	56	count
loop_edge_share	0.1409	share
edge_count	1398	count

Canalicular width, the new measure

measure	value	units
width_mean_px	8.2307	px
width_median_px	8.2462	px
width_p10_px	4	px
width_p90_px	11.6619	px
width_pixels_used	23613	count

Canalicular width: 2 x the Euclidean distance transform of the canalicular mask at each skeleton pixel. Measured only on skeleton pixels on the canalicular mask: the lacuna buffer and the dilated vascular mask are excluded. The threads are about 2 to 6 px wide, so the width is close to the pixel grid and quantised in steps of about 1 px, it moves with the threshold (a lower cut widens every thread), and it has not been validated against any manual measurement.

Lacuna shape

lacuna_id	area_px2	major_axis_length_px	minor_axis_length_px	aspect_ratio	eccentricity	solidity	orientation_rad	centroid_row_px	centroid_col_px	on_border
1	5018	113.612	66.0072	1.7212	0.8139	0.7134	-1.5671	33.3908	243.731	True
2	1429	75.9455	34.6321	2.1929	0.89	0.5788	0.9821	47.7124	162.336	False
3	12478	404.05	54.0339	7.4777	0.991	0.6138	1.5354	38.8529	844.287	True
4	1943	78.2261	36.7629	2.1279	0.8827	0.7692	-1.4856	56.1004	636.36	False
5	1748	63.7004	52.4126	1.2154	0.5683	0.625	1.3536	36.7677	399.808	False
6	482	37.8235	20.2932	1.8638	0.8439	0.7226	-0.6658	78.6515	500.948	False
7	2001	74.781	36.9486	2.0239	0.8694	0.794	-1.1924	101.66	153.378	False
8	4592	90.8676	67.3599	1.349	0.6712	0.8423	-1.2953	137.617	765.059	False
9	902	70.1443	23.2471	3.0173	0.9435	0.6677	-1.2472	150.552	535.413	False
10	18263	251.192	114.111	2.2013	0.8909	0.7282	-1.0667	218.389	97.7195	True
11	2333	67.9441	45.1639	1.5044	0.7471	0.9554	-0.4874	234.38	1000.91	True
12	11887	168.982	95.145	1.7761	0.8264	0.8606	-1.2985	286.803	681.158	False
13	697	50.8222	20.9969	2.4205	0.9107	0.7383	1.293	287.073	212.017	False
14	3532	87.4182	55.1855	1.5841	0.7756	0.7638	1.235	329.842	344.026	False
15	4129	107.816	50.0459	2.1543	0.8857	0.9077	-1.34	378.661	498.241	False
16	889	60.4502	27.413	2.2052	0.8913	0.6247	-1.2388	363.547	49.8526	False
17	514	40.4388	21.1806	1.9092	0.8519	0.6993	0.9864	406.87	807.179	False
18	587	53.646	23.5495	2.278	0.8985	0.6501	-0.399	433.826	208.523	False
19	1735	68.9793	37.2098	1.8538	0.842	0.6893	-0.903	463.884	594.474	False
20	751	42.1527	25.0435	1.6832	0.8044	0.8632	-1.1817	459.581	482.401	False
21	508	31.1659	30.9736	1.0062	0.1109	0.6737	0.679	475.947	802.87	False
22	5417	114.374	70.6956	1.6178	0.7861	0.6168	1.316	527.301	55.7192	True
23	3037	99.839	58.2931	1.7127	0.8118	0.571	1.4007	558.019	301.395	False
24	4073	115.2	49.8465	2.3111	0.9015	0.7391	0.3504	603.966	867.145	False

25	2337	72.0848	52.1015	1.3835	0.6911	0.6974	-1.4357	607.677	651.435	False
26	1679	74.0209	37.2141	1.9891	0.8644	0.724	-0.9638	634.934	701.784	False
27	7591	130.597	95.8386	1.3627	0.6793	0.6789	-0.2363	683.477	82.1523	False
28	4140	142.999	43.9592	3.253	0.9516	0.7236	-1.1238	670.561	384.423	False
29	554	30.7716	24.9383	1.2339	0.5858	0.895	1.0673	729.574	697.603	False
30	549	48.7614	15.8873	3.0692	0.9454	0.8356	-0.9101	738.328	786.444	False
31	640	57.661	17.8443	3.2313	0.9509	0.7018	-0.3969	793.93	358.258	False
32	2036	78.7104	35.4441	2.2207	0.8929	0.8879	-1.002	828.682	94.3777	False
33	45573	496.877	154.577	3.2144	0.9504	0.7075	1.5056	966.648	733.764	True
34	1332	74.3708	26.5147	2.8049	0.9343	0.75	-1.2358	829.074	388.601	False
35	6240	115.676	84.3355	1.3716	0.6844	0.7226	-1.0938	881.946	274.251	False
36	1277	51.634	38.2716	1.3491	0.6713	0.7854	1.2894	872.15	1003.8	True

Network per lacuna

lacuna_id	roots_count	ring_length_r30_px	ring_length_r60_px	owned_length_px	edge_count	mean_edge_length_px	ring_width_mean_r30_px
1	15	337	427	568.174	29	19.5922	6.7223
2	12	274	358	364.877	23	15.8642	6.4715
3	24	502	617	535.973	32	16.7491	6.8655
4	10	293	534	426.534	28	15.2334	7.2329
5	9	352	485	357.208	22	16.2367	7.3539
6	11	264	366	438.534	27	16.242	7.5091
7	7	240	321	161.439	10	16.1439	6.8457
8	15	432	634	749.227	38	19.7165	7.593
9	8	315	698	776.683	38	20.439	7.6403
10	18	515	788	919.666	48	19.1597	7.5529
11	6	155	358	401.421	19	21.1274	8.7355
12	26	659	1175	1353.12	67	20.1958	8.1509
13	7	204	384	431.333	23	18.7536	8.7157
14	16	500	1052	1999	93	21.4946	8.5149
15	11	341	658	1054.25	55	19.1681	8.3777
16	7	179	235	246.492	10	24.6492	6.1524
17	7	218	484	1118.03	52	21.5006	7.5444
18	6	230	501	349.907	23	15.2133	8.3811
19	12	360	668	1001.49	35	28.614	7.7108
20	5	155	258	175.125	10	17.5125	8.2917
21	6	248	432	392.049	18	21.7805	6.9808
22	19	351	508	724.855	34	21.3193	5.9354
23	19	379	844	1380.11	69	20.0016	7.8666
24	15	420	955	815.21	40	20.3803	7.0277
25	15	277	585	667.245	36	18.5346	7.6247
26	9	167	339	207.48	7	29.64	7.1073
27	23	499	869	903.436	46	19.6399	7.5211
28	9	207	349	267.61	17	15.7418	7.6147
29	5	168	322	118.498	6	19.7496	8.2887
30	4	177	394	201.723	5	40.3446	8.1657
31	1	35	62	7.8284	1	7.8284	12.9591
32	8	232	483	506.031	22	23.0014	8.1991
33	39	777	1166	1686.33	88	19.1629	8.6448
34	6	119	213	157.468	10	15.7468	7.2164
35	10	227	419	323.622	15	21.5748	8.0692
36	6	145	428	660.345	36	18.3429	10.4527

Normalised per lacuna

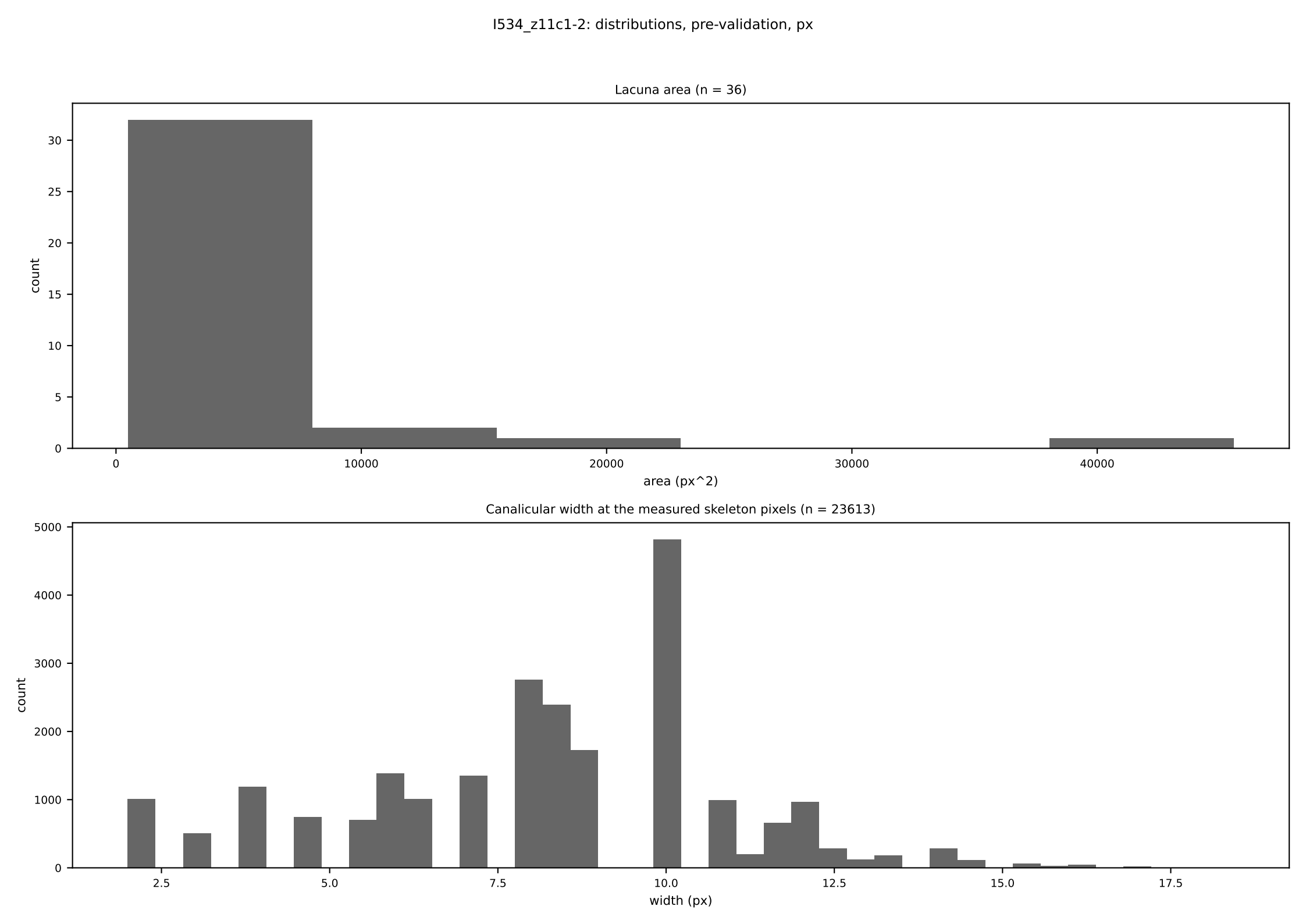
lacuna_id	perimeter_px	ring_area_r30_px2	ring_area_r60_px2	in_frame_fraction_r30	in_frame_fraction_r60	ring_density_r30	ring_density_r60	roots_per_100px_perimeter
1	494.351	8231	11715	0.7421	0.6708	0.0409428	0.036449	3.0343

2	282.291	4862	7142	0.9455	0.8134	0.0563554	0.050126	4.2509
3	1255.09	18964	24340	0.693	0.5601	0.0264712	0.0253492	1.9122
4	253.279	7100	12097	1	0.9073	0.0412676	0.0441432	3.9482
5	270.635	8155	13291	0.8695	0.7737	0.0431637	0.0364909	3.3255
6	123.054	5704	10031	1	1	0.0462833	0.0364869	8.9392
7	236.936	5018	6853	1	1	0.0478278	0.0468408	2.9544
8	341.505	10661	17446	1	1	0.0405215	0.0363407	4.3923
9	176.167	7534	16220	1	1	0.0418105	0.0430333	4.5412
10	1221.79	20165	28289	0.8577	0.7956	0.0255393	0.0278554	1.4732
11	199.782	5730	14247	0.6804	0.6335	0.0270506	0.0251281	3.0033
12	548.257	17191	32458	1	1	0.038334	0.0362006	4.7423
13	136.125	6658	12133	1	1	0.0306398	0.0316492	5.1423
14	297.706	11030	23116	1	1	0.0453309	0.0455096	5.3744
15	305.605	10599	17028	1	1	0.0321729	0.0386422	3.5994
16	191.167	7042	12138	0.9189	0.7962	0.0254189	0.0193607	3.6617
17	132.267	5680	13004	1	1	0.0383803	0.0372193	5.2923
18	139.267	6658	17676	1	1	0.0345449	0.0283435	4.3083
19	238.451	9266	19320	1	1	0.0388517	0.0345756	5.0325
20	122.953	5564	11136	1	1	0.0278577	0.0231681	4.0666
21	122.953	5620	11532	1	1	0.0441281	0.037461	4.8799
22	490.399	11727	20264	0.8061	0.7376	0.0299309	0.0250691	3.8744
23	378.475	12733	26807	1	1	0.0297652	0.0314843	5.0201
24	373.546	12922	27828	1	1	0.0325027	0.0343179	4.0156
25	290.877	7526	16187	1	1	0.0368057	0.0361401	5.1568
26	233.38	6410	12080	1	1	0.026053	0.0280629	3.8564
27	555.955	15261	25089	0.9401	0.8614	0.0326977	0.0346367	4.137
28	383.889	12892	25468	1	1	0.0160565	0.0137035	2.3444
29	97.9828	5438	10968	1	1	0.0308937	0.0293581	5.1029
30	115.539	5958	13707	1	1	0.029708	0.0287444	3.462
31	139.054	4710	7157	1	1	0.007431	0.00866285	0.7191
32	222.693	8786	20073	1	0.9917	0.0264056	0.0240622	3.5924
33	2556.07	33726	52704	0.6475	0.572	0.0230386	0.0221236	1.5258
34	196.066	6012	12715	1	1	0.0197937	0.0167519	3.0602
35	446.416	13647	28597	1	1	0.0166337	0.0146519	2.2401
36	206.752	4961	9598	0.6755	0.6216	0.029228	0.0445926	2.902

Network v2 per lacuna

lacuna_id	ring_attached_length_r30_px	ring_attached_length_r60_px	ring_length_w_r30_px	ring_length_w_r60_px	sholl_crossings_r10	sholl_crossings_r20	sholl_crossings_r30
1	334	421	381.806	480.818	9	8	4
2	271	315	309.35	406.605	9	6	3
3	438	497	563.387	698.583	16	12	5
4	201	323	331.421	601.588	5	6	12
5	348	425	392.836	534.433	6	9	5
6	253	314	295.794	412.877	5	5	5
7	151	151	275.522	367.291	5	4	4
8	413	591	496.446	725.441	11	11	12
9	278	600	361.149	803.311	6	10	9
10	465	540	581.902	895.394	12	14	9
11	140	252	165.669	393.836	3	4	4
12	639	1127	726.487	1314.56	15	15	17
13	196	285	235.38	437.676	5	6	4
14	486	1022	568.688	1196.76	10	16	18
15	327	612	395.505	761.553	8	9	12
16	153	156	207.794	271.321	5	2	3
17	184	413	243.924	553.274	5	8	6
18	180	298	267.865	576.529	3	6	8
19	340	593	412.605	776.039	6	7	9

20	145	178	175.297	287.551	4	4	4
21	221	303	279.966	494.931	5	5	8
22	327	352	403.576	592.541	7	3	5
23	370	705	430.818	954.365	11	10	10
24	408	696	481.203	1073.34	8	12	12
25	232	384	316.179	665.186	3	6	7
26	153	183	181.853	382.019	6	3	3
27	451	748	578.399	1005.9	13	14	13
28	199	281	226.853	381.007	5	4	7
29	149	248	184.054	363.978	4	6	4
30	115	165	202.995	451.789	4	7	7
31	10	10	37.799	67.6985	0	1	2
32	232	429	267.794	549.902	6	6	7
33	716	1035	866.323	1308.97	31	16	9
34	119	159	135.397	236.238	2	2	3
35	181	252	251.137	462.576	7	7	9
36	139	405	158.468	481.161	5	4	4


I534_z11c1-2: distributions, pre-validation, px
Lacuna area (n = 36)
count
area (px^2)
0
10000
20000
30000
40000
Canalicular width at the measured skeleton pixels (n = 23613)
count
width (px)
2.5
5.0
7.5
10.0
12.5
15.0
17.5